CROA: CROSS-REPRESENTATION OPTIMAL ALIGNMENT FOR FINE-GRAINED TARGETED TRANSFER ATTACKS ON MLLMS

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ABSTRACT

Multimodal large language models (MLLMs) are exposed to targeted transfer attacks that steer a black-box victim toward attacker-chosen semantics using perturbations optimized on surrogate encoders. For *fine-grained* targeting, existing attacks predominantly draw their surrogates from the image–text contrastive paradigm, which focuses on high-level semantic categories but may under-constrain fine-grained spatial details. We propose CROA (Cross-Representation Optimal Alignment), integrating contrastive and self-supervised representations where the two paradigms provide complementary local visual information. CROA couples a multi-architecture CLIP ensemble with DINOv2 by establishing clustered local optimal transport across patch tokens in both feature spaces alongside global feature alignment. It further incorporates source repulsion and multi-crop momentum optimization under a unified configuration. Extensive experiments across six commercial and open-source MLLM victims show that CROA achieves the highest mean ASR among the evaluated methods under our benchmark settings, attaining 51.0% on commercial frontier models and 92.0% on open-source models, outperforming both single- and multi-paradigm baselines. Code is available at https://github.com/chenwudao/CROA.

***Index Terms*—** Adversarial attack, targeted transfer attack, multimodal language model, optimal transport, fine-grained targeting

1. INTRODUCTION

Multimodal large language models (MLLMs) [1, 2] have achieved remarkable progress in visual understanding and natural language generation. As MLLMs are increasingly deployed in real-world applications, their vulnerability to adversarial examples has become a pressing security concern [3]. Among various attack surfaces, *targeted transfer attacks* are particularly practical: perturbations are optimized on locally available surrogate encoders and transferred to black-box victims with zero queries, steering the victim's caption toward an attacker-chosen target image [4, 5].

Along this line, AttackVLM [3] aligns projected CLIP features via ensemble cosine similarity. ITA [6] investigates imperceptible transfer attacks under perturbation constraints. M-Attack [4] introduces random-crop augmentation on CLS embeddings, showing that input diversity enhances transferability. FOA-Attack [5] couples global CLS cosine alignment with local optimal transport (OT) on clustered patch tokens to match feature distributions within contrastive feature space. Despite this algorithmic progression, these methods remain confined to the image–text contrastive paradigm. Contrastive pretraining optimizes cross-modal alignment, which captures high-level semantics effectively but may provide limited localized visual structure. In contrast, self-supervised representations provide rich, complementary local patch features.

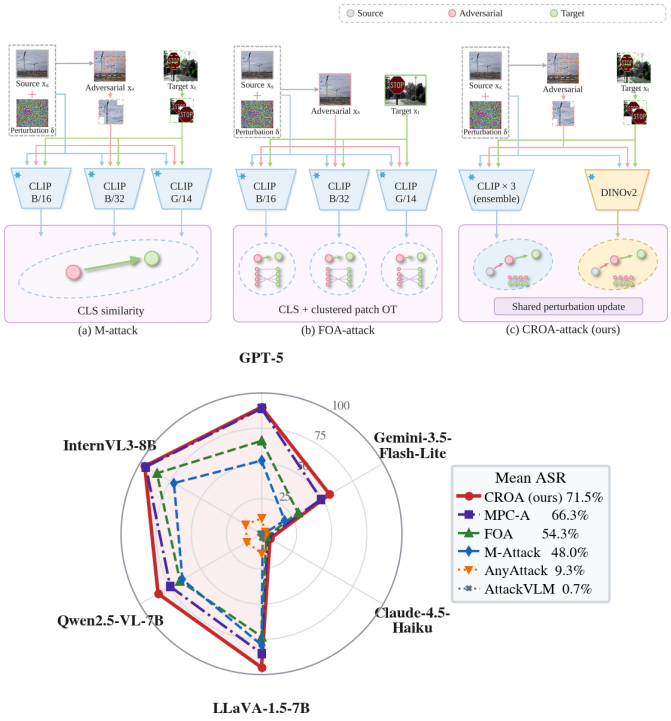


Fig. 1. From single-paradigm to cross-representation alignment. Top: (a) M-Attack, (b) FOA, and (c) CROA architectures. Bottom: Targeted ASR (%) across six commercial and open-source victims, with aggregate 6-victim mean in the legend.

To leverage multiple paradigms, MPCAttack [14] combines CLIP, DINOv2, and an auxiliary MLLM surrogate with a source-repulsion term. However, it relies on *coarse global pooling*, concatenating 1D global vectors (such as DINOv2 CLS tokens) into a joint similarity objective while requiring target text prompts and white-box MLLM queries during optimization. By discarding dense patch tokens in the self-supervised branch, it cannot exploit the complementary spatial and structural details preserved in DINOv2 patch fields. Under the tight L_∞ budget, residual source features can also persist, leading to "double exposure" captions that conflate source and target concepts.

To address these limitations, we propose Cross-Representation Optimal Alignment (CROA). Unlike prior works that perform patch-level optimal transport only within contrastive models [5]

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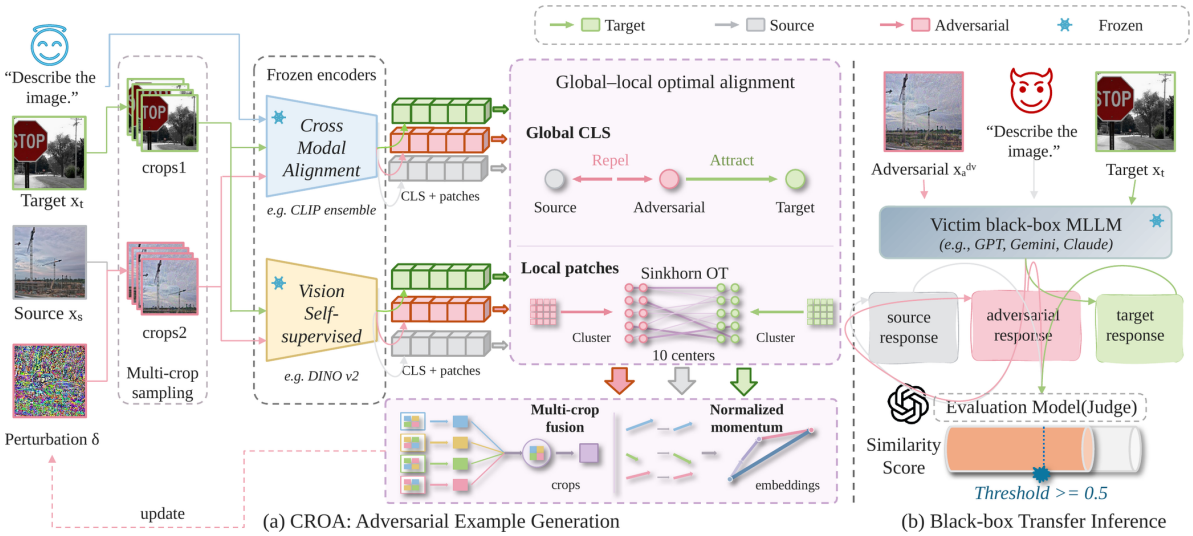


Fig. 2. CROA overview. Frozen CLIP and DINOv2 encoders provide global and local target-alignment objectives. Global source repulsion complements target attraction; clustered patch OT supplies local matching. Four-view averaging and normalized momentum update a shared perturbation. Victim captions and target references are compared only at evaluation time.

or collapse self-supervised representations to global pooled vectors [14], CROA establishes fine-grained local optimal transport across patch tokens in *both* CLIP and DINOv2 feature spaces. In a vision-only setting without target text or MLLM surrogates, CROA couples global alignment with clustered local OT, incorporates cross-representation source repulsion, and stabilizes updates with multi-crop normalized momentum.

Our main contributions are summarized as follows:

- We propose a vision-only cross-representation attack framework that performs local optimal transport across both contrastive and self-supervised patch spaces alongside global alignment and source repulsion, generating transferable perturbations without target text or MLLM surrogates.
- We evaluate CROA across three commercial frontier MLLMs (GPT-5, Gemini-3.5-FL, Claude-4.5-H) and three open-source MLLMs (LLaVA-1.5, Qwen2.5-VL, InternVL3), achieving top mean ASR on both commercial (51.0%) and open-source models (92.0%, exceeding MPC-A by 7.0 points), with an overall 71.5% mean across all six victims.
- We provide component ablations isolating representation paradigm, local OT, and source repulsion across all six architectures, demonstrating their contributions to aggregate attack success.

2. METHODOLOGY

2.1. Preliminaries

Let $\mathcal{V}$ denote a black-box victim MLLM that generates caption $y = \mathcal{V}(x, p)$ given an input image x and a fixed captioning prompt p ("Describe this image in one concise sentence, no longer than 20 words."). Given a source image x_s and an attacker-chosen target image x_t whose target caption is $y_t = \mathcal{V}(x_t, p)$, the adversary seeks an adversarial perturbation δ bounded by $\|\delta\|_\infty \le \varepsilon$ ($\varepsilon = 16/255$) to steer the victim's caption toward the target concept:

$$\max_{\|\delta\|_\infty \le \varepsilon} \mathcal{S}\left(\mathcal{V}(\mathrm{clip}_{[0,1]}(x_s + \delta), p),\ y_t\right), \tag{1}$$

Table 1. Surrogate interfaces. DINOv2-B/14 exposes the same CLS/patch-token interface as the contrastive extractors.

Extractor	Paradigm	CLS dim	Patches
CLIP-B/16	contrastive	768	196×768
CLIP-B/32	contrastive	768	49×768
LAION-G/14	contrastive	1408	256×1408
DINOv2-B/14	self-sup.	768	256×768

where $\mathcal{S}(\cdot,\cdot)$ measures semantic similarity via an LLM judge. Because $\mathcal{V}$ is query-free and gradient-inaccessible, direct optimization of Eq. (1) is impossible. Instead, the adversary optimizes an adversarial surrogate objective $\mathcal{L}_{\mathrm{CROA}}$ over a local surrogate ensemble $\mathcal{F}$:

$$\delta^* = \arg\max_{\|\delta\|_\infty \le \varepsilon} \mathcal{L}_{\mathrm{CROA}}(x_s + \delta,\ x_t,\ x_s;\ \mathcal{F}), \tag{2}$$

and directly transfers $x = \mathrm{clip}_{[0,1]}(x_s + \delta^*)$ to $\mathcal{V}$ with zero victim queries. Fig. 2 illustrates the overall framework.

2.2. Cross-paradigm surrogate fusion

Fig. 2(a) illustrates the surrogate pipeline. The contrastive branch $\mathcal{C}$ comprises CLIP-B/16, CLIP-B/32, and LAION CLIP-G/14 [9, 10]; the self-supervised branch d is DINOv2-B/14 [15]. For every encoder k, we extract the final-layer CLS token $f_k^{\mathrm{cls}}(x)$ and patch tokens $\{f_{k,\ell}^{\mathrm{tok}}(x)\}_{\ell=1}^{N_k}$, ℓ_2-normalizing each:

$$c_k(x) = \frac{f_k^{\mathrm{cls}}(x)}{\|f_k^{\mathrm{cls}}(x)\|_2}, \quad p_{k,\ell}(x) = \frac{f_{k,\ell}^{\mathrm{tok}}(x)}{\|f_{k,\ell}^{\mathrm{tok}}(x)\|_2}. \tag{3}$$

While CLIP aligns images with natural language to provide high-level semantic guidance, DINOv2 provides complementary local visual features capturing spatial layout and object structure. Crucially, unlike MPCAttack [14], which collapses DINOv2 into a single CLS token and relies on target text guidance alongside an MLLM surrogate, CROA preserves full patch token fields in both paradigms and performs fine-grained local optimal transport within their native metric spaces in a purely vision-to-vision setting.

Table 2. Targeted ASR (%) on three commercial victims. Bold marks column maxima.

Method	GPT-5		Gemini-3.5-FL		Claude-4.5-H		Mean	
	ASR	KMR	ASR	KMR	ASR	KMR	ASR	KMR
AttackVLM [3]	0.0	6.0	0.0	4.2	0.0	4.1	0.0	4.7
AnyAttack [11]	11.0	14.8	3.0	9.1	3.0	**10.5**	5.7	11.5
M-Attack [4]	52.0	49.1	19.0	28.6	1.0	7.4	24.0	28.4
FOA [5]	66.0	51.8	30.0	32.2	4.0	9.8	33.3	31.3
MPC-A [14]	89.0	65.0	49.0	43.2	5.0	10.2	47.7	39.5
CROA	**90.0**	**65.1**	**56.0**	**49.3**	**7.0**	9.5	**51.0**	**41.3**

2.3. Source-aware global–local OT alignment

Global attraction and repulsion. As shown in Fig. 2(b), optimizing solely toward the target without penalizing source similarity can leave residual source patterns under a tight L_∞ budget, inducing double exposure. We thus incorporate global attraction and source repulsion in each encoder space:

$$G_k(x) = c_k(x)^\top c_k(x_t) - \beta\, c_k(x)^\top c_k(x_s), \tag{4}$$

where $\beta = 0.5$ is shared across both branches; the second term expels residual source features from both contrastive ($\mathcal{C}$) and self-supervised (d) representation spaces. Source references $\{c_k(x_s)\}$ are precomputed once from the clean image.

Local clustered transport. Following FOA's formulation [5], we match patch token distributions via optimal transport. Patch tokens of each crop are assigned to $K_c = 10$ clusters by iterative maximum-dot-product assignment; each centroid $a_i = |S_i|^{-1} \sum_{p \in S_i} p$ is a token mean, with $B = \{b_j\}_{j=1}^{K_c}$ obtained analogously for the target. On the similarity matrix $S_{ij} = a_i^\top b_j$, we run entropic Sinkhorn iterations [12, 13] on the Gibbs kernel $K_{ij} = \exp[(S_{ij} - 1)/0.1]$ under uniform marginals:

$$u \leftarrow \frac{\mathbf{1}/K_c}{Kv}, \qquad v \leftarrow \frac{\mathbf{1}/K_c}{K^\top u}, \tag{5}$$

for at most 100 iterations or until convergence ($< 10^{-2}$). The resulting plan $\mathbf{P}^* = \mathrm{diag}(u) K\, \mathrm{diag}(v)$ is detached from the graph, and the local objective transports feature content through:

$$T_k(x) = \sum_{i=1}^{K_c} \sum_{j=1}^{K_c} \mathbf{P}^*_{ij}\, a_i^\top b_j. \tag{6}$$

Stop-gradients through $\mathbf{P}^*$ and discrete assignments allow gradients to flow directly through the centroid means.

Cross-representation objective. With $L_k = G_k + \lambda_k T_k$, where $\lambda_C = 0.2$ for CLIP and $\lambda_D = 0.4$ for DINOv2, the fused objective balances both paradigms:

$$L_{\mathrm{CROA}} = \frac{0.5}{|\mathcal{C}|} \sum_{k \in \mathcal{C}} L_k + 0.5\, L_d. \tag{7}$$

The three CLIP encoders share half the objective and DINOv2 receives the other half, preventing encoder count from dominating either paradigm. Furthermore, DINOv2 employs a stronger local weight ($\lambda_D = 0.4$ vs. $\lambda_C = 0.2$) because self-supervised tokens provide complementary local visual features that capture localized structure, tolerating richer local transport without corrupting global semantics.

2.4. Multi-crop momentum optimization

At each step, we sample $M = 4$ independent source and target views with crop area fractions in $[0.35, 1.0]$ to capture multi-scale contexts. Unlike unnormalized momentum in prior attacks [14], we average multi-crop losses and apply mean-absolute gradient normalization [8]:

$$\begin{aligned} L &= M^{-1} \sum_{m=1}^{M} L_{\mathrm{CROA}}^{(m)}, \\ m_{t+1} &= \mu\, m_t + \frac{\nabla_\delta L}{\mathrm{mean}\,|\nabla_\delta L| + 10^{-8}}, \\ \delta_{t+1} &= \Pi_{\|\delta\|_\infty \le \varepsilon} \left(\delta_t + \alpha\ \mathrm{sign}(m_{t+1})\right), \end{aligned} \tag{8}$$

with $\mu = 0.9$, $\alpha = 1/255$, $\delta_0 = m_0 = 0$, and $T = 300$ steps. Mean normalization decouples step magnitude from loss-scale differences across paradigms, stabilizing gradient stochasticity from multi-crop sampling and patch clustering. All hyperparameters are shared across victims without any per-victim tuning.

3. EXPERIMENTS

3.1. Experimental settings

Baselines. We compare CROA with representative transferable adversarial attack methods on MLLMs, including AttackVLM [3], AnyAttack [11], M-Attack [4], FOA [5], and MPC-A [14]. Baselines are reproduced following their open-source implementations, adapting text inputs where required (such as target descriptions for MPC-A), under unified perturbation budgets, test samples, and evaluation pipelines.

Datasets and victims. Following the standard benchmark protocols of M-Attack [4] and FOA [5], we adopt the 100 source–target image pairs constructed from the NIPS 2017 Adversarial Attacks and Defenses Competition dataset along with their fine-grained target keyword annotations to ensure a fair and consistent comparison. Evaluated victims encompass both representative closed-source commercial models (GPT-5 [16], Gemini-3.5-Flash-Lite [17], and Claude-4.5-Haiku [18]) and popular open-source MLLMs (LLaVA-1.5-7B [19], Qwen2.5-VL-7B [20], and InternVL3-8B [21]).

Attack configurations. Following standard practice [5, 14], all iterative attacks run for $T = 300$ steps under an L_∞ perturbation budget of $\varepsilon = 16/255$ with step size $\alpha = 1/255$. For CROA, hyperparameters are set to $\beta = 0.5$, $\lambda_C = 0.2$, $\lambda_D = 0.4$, $w_C = w_D = 0.5$, $M = 4$ crops in $[0.35, 1.0]$, and momentum $\mu = 0.9$, kept strictly frozen across all victims. All perturbation generation runs on a single NVIDIA RTX 6000 Ada (48 GB) GPU.

Evaluation metrics. Following the evaluation protocol of FOA [5], we adopt the LLM-as-a-judge framework using GPT-4o-mini [22]. We evaluate two primary metrics: (1) **Targeted ASR**:

Table 3. Open-source ASR (%). A three-model mean is reported across all evaluated architectures (100 pairs each).

Method	LLaVA	Qwen2.5	InternVL3	Mean
AttackVLM [3]	3.0	0.0	1.0	1.3
AnyAttack [11]	14.0	12.0	13.0	13.0
M-Attack [4]	79.0	65.0	72.0	72.0
FOA [5]	73.0	67.0	86.0	75.3
MPC-A [14]	85.0	75.0	95.0	85.0
CROA	**95.0**	**85.0**	**96.0**	**92.0**

Table 4. Component analysis: ASR (%) across all six commercial and open-source victims (100 pairs each). Bold marks column maxima.

Variant	GPT-5	Gemini	Claude	LLaVA	Qwen	IVL3	Mean
FOA (CLIP)	66	30	4	73	67	86	54.33
CLIP + Multi-scale	83	48	5	85	76	92	64.83
CROA w/o DINOv2	89	56	**9**	92	79	88	68.83
CROA w/o local OT	88	**60**	6	89	81	95	69.83
CROA w/o repulsion	85	57	4	91	79	95	68.50
Full CROA	**90**	56	7	**95**	**85**	**96**	**71.50**

following FOA's prompting format inspired by GPTScore [7], GPT-4o-mini directly scores the semantic similarity between the victim caption and target description; an attack succeeds if the score is ≥ 0.5. (2) **Keyword Match Rate (KMR)**: GPT-4o-mini evaluates the recall of target-specific keywords within the victim caption, measuring fine-grained semantic transfer fidelity.

3.2. Main results

Table 2 reports targeted attack results on three commercial victims. CROA attains the highest aggregate ASR (51.0%) and KMR (41.3%) among evaluated baselines under our benchmark settings, outperforming MPC-A (47.7% ASR, 39.5% KMR) and FOA (33.3% ASR, 31.3% KMR). This demonstrates that jointly aligning contrastive semantics and dense self-supervised geometry substantially enhances cross-model transferability in a vision-only setting. On GPT-5 and Gemini-3.5-FL, CROA achieves 90.0% and 56.0% ASR, improving over FOA by 24.0 and 26.0 percentage points, respectively. While Claude-4.5-H remains challenging across all evaluated attacks, CROA still achieves the highest ASR of 7.0%. Furthermore, CROA demonstrates superior semantic fidelity, reaching 65.1% KMR on GPT-5 and 49.3% on Gemini.

The open-source panel (Table 3) gives CROA a 92.0% mean ASR, versus 85.0% for MPC-A and 75.3% for FOA, achieving 95.0% on LLaVA, 85.0% on Qwen2.5-VL, and 96.0% on InternVL3. A paired exact McNemar test confirms the significance of CROA's gain over FOA on LLaVA ($p < 0.0001$, 23 wins vs. 1 loss; $p = 0.021$ vs. MPC-A). Across these diverse architectures, the frozen surrogate ensemble transfers effectively without victim-specific tuning.

3.3. Ablation study

Representation complementarity. Table 4 evaluates each component across all six target architectures. Compared with the CLIP+Multi-scale baseline, full CROA improves mean ASR by 6.67 points (64.83% to 71.50%), reaching consistent gains across victims. Ablating DINOv2 reduces mean ASR to 68.83%, with

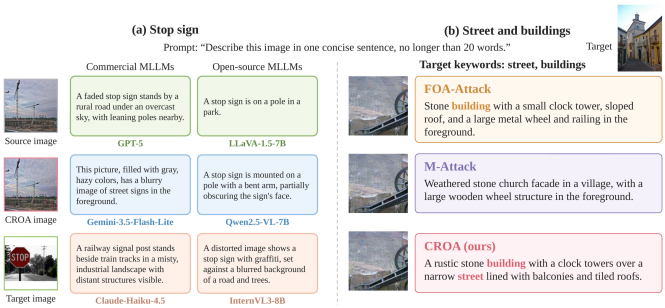


Fig. 3. Selected target-detail examples. Left: CROA captions from six victims. Right: the CROA caption includes both "street" and "building," illustrating keyword coverage beyond the main object.

pronounced drops on modern dense-patch open-source models: InternVL3 falls by 8 points (96% to 88%) and Qwen2.5 falls by 6 points (85% to 79%), alongside target-detail KMR gains from DINOv2 (e.g., GPT-5 KMR: 61.4% to 65.1%; LLaVA KMR: 64.0% to 67.4%). While Claude marginally favors CLIP alone (9% vs. 7%), DINOv2 provides crucial geometric transferability on vision-centric backbones.

Local matching and source awareness. Removing local OT reduces mean ASR to 69.83%, triggering a six-point drop on LLaVA (95% to 89%) and a four-point drop on Qwen2.5 (85% to 81%), confirming the value of patch-level distribution alignment. Removing source repulsion incurs the heaviest penalties on high-capacity models: GPT-5 drops by 5 points (90% to 85%), Qwen2.5 by 6 points (85% to 79%), and LLaVA by 4 points (95% to 91%), reducing mean ASR to 68.50%. Full CROA achieves the top mean ASR (71.50%) and dominates on four of six victims, demonstrating that dual-paradigm alignment, fine-grained local transport, and active source repulsion collectively contribute to overall transferability.

3.4. Qualitative results

Figure 3 illustrates qualitative results across diverse MLLMs. On the left, CROA steers most victims toward target-aligned descriptions (e.g., a stop sign or signal post) while reflecting architectural variations. On the right, CROA induces detailed captions covering fine-grained context such as "street" and "building", demonstrating effective target-detail semantic coverage.

4. CONCLUSION

We investigated whether self-supervised visual representations complement image–text contrastive features in fine-grained targeted transfer attacks. CROA combines global feature alignment and clustered local optimal transport across CLIP and DINOv2 representations, with source repulsion and multi-crop momentum optimization. On 100 source–target pairs, CROA achieves mean attack success rates of 51.0% across three commercial victims and 92.0% across three open-source victims, the highest among the baselines evaluated under our protocol. Its mean target-keyword recall on the commercial victims is 41.3%. Component ablations show that DINOv2, local transport, and source repulsion each improve aggregate attack success, although their effects vary across victims. These results support cross-representation alignment for vision-only perturbation generation without target descriptions or an MLLM surrogate.

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